



ILUMYS[®]

INTENSIVE ANTI-DARK SPOT

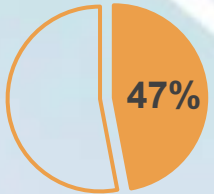
SKIN PERFECTOR & ILLUMINATOR

ILUMYS[®]

INTENSIVE ANTI-DARK SPOT

GREEN TECH

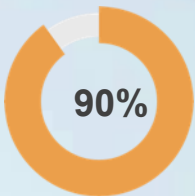
UNCOVERING SKIN EVENNESS WORLDWIDE CONTEXT



People feel uncomfortable with their skin problem

40⁺
S'

Common in people over 40



Of people over 70

Sunlight
Pollution
Chemical
Medication



EXTERNAL

Age
Hormonal
Inflammatory
Illness



INTERNAL

SCIENTIFIC CONTEXT IN THE CONTINUITY OF SKIN HEALTHSPAN

Dark spots arise from imbalance of melanin regulation and lipofuscin accumulation

Solution :
target both lipofuscin and
melanin accumulation
as well as implicated
surrounding skin cells



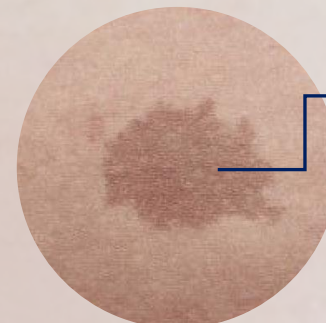
Beige, light brown color

Formed with age and
accumulation of waste
material within cells



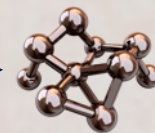
Lipofuscin

- Made of oxidized proteins and lipids
- Formed in  **Lysosomes**
- Waste material 



Dark brown color

Formed with exposition to
UV light, which triggers
melanin production



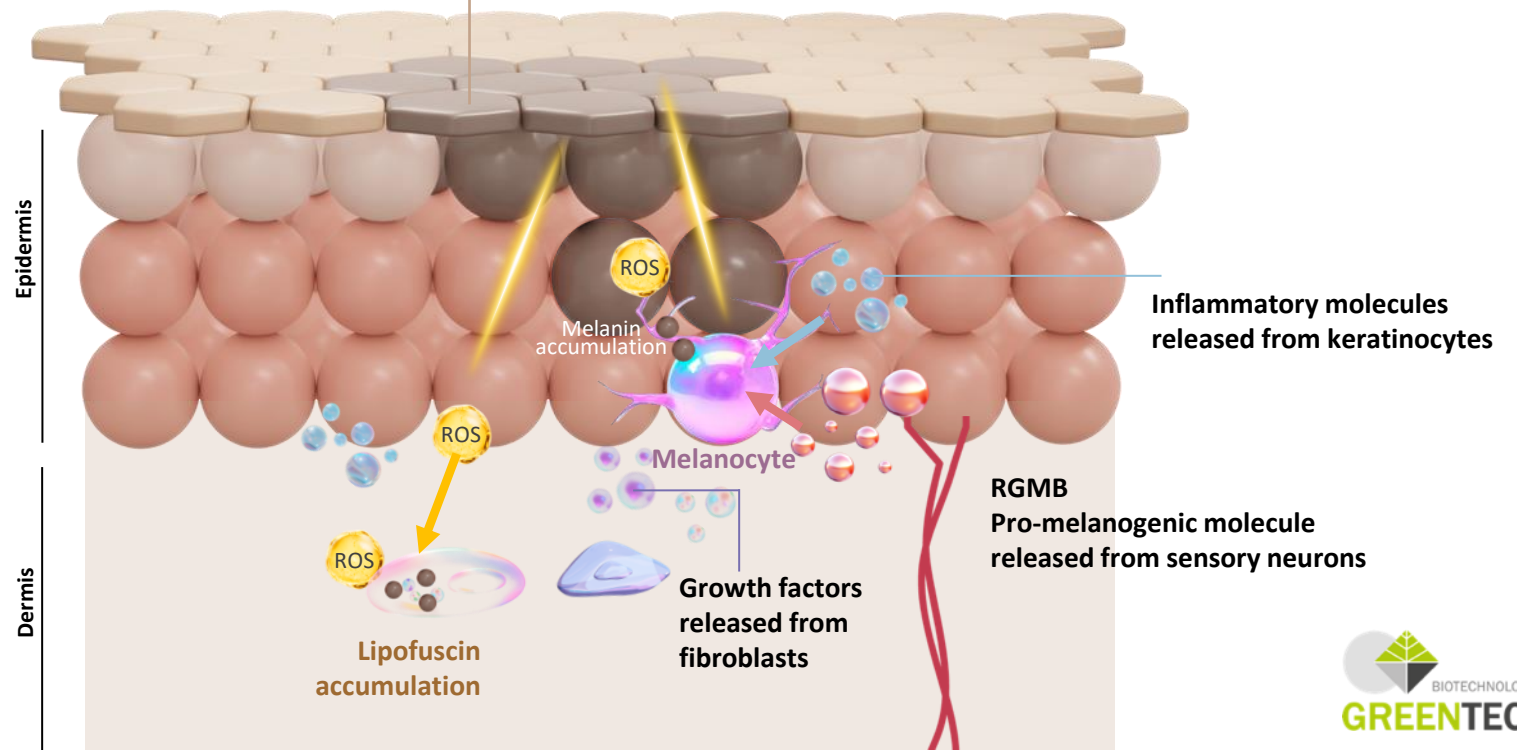
Melanin

- Made of eumelanin and phenomelanin
- Produced by  **melanocytes**
- Skin protection 

ROOT CAUSE

DARK SPOT MECHANISM

DARK SPOT / AGE SPOT



DUAL ACTION

LIPOFUSCIN ACCUMULATION AND MELANOGENESIS DYSREGULATION

GREENTECH INNOVATION



POWERFUL BIOACTIVE MOLECULES
GINGEROLS FROM GINGER

ORIGIN & DEVELOPMENT

Skin tone balancing spot treatment
SOURCED FROM
GINGER ROOTS (*Zingiber officinale*)

ASIAN POTENT TRADITIONAL MEDECINES
Used in Asian pharmacopoeia for thousands of years



LIGHTENING
PHYTOMOLECULES:
GINGEROLS

TITRATION

- Total gingerols: > 32%/DM
- 6-gingerol: > 20%/DM
- 8-gingerol: > 4%/DM
- 10-gingerol: > 4%/DM

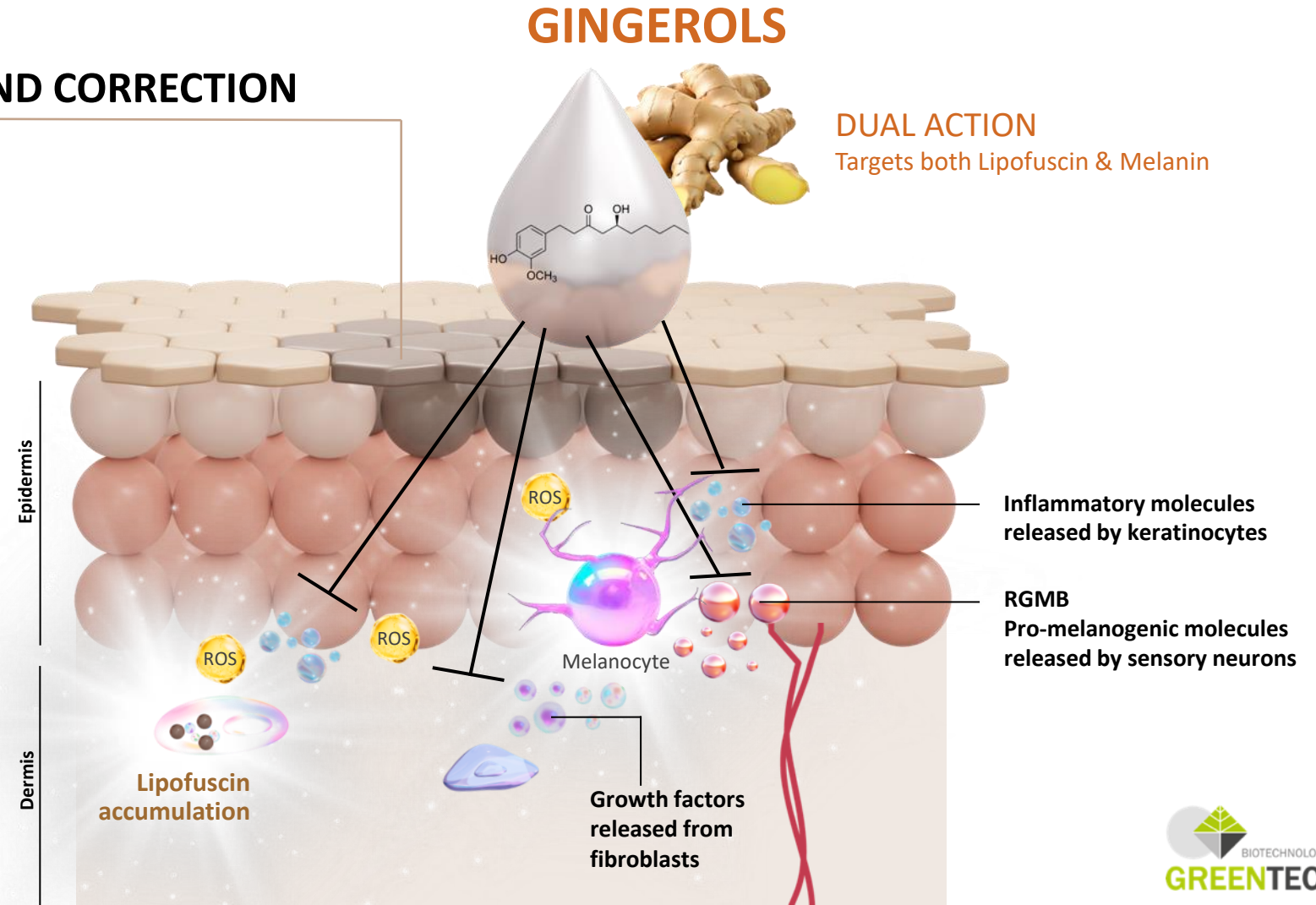
GINGEROLS

Known for their antioxidant and soothing properties
6-gingerol: longevity and stress-resistant properties

Lee et al., 2018

MECHANISM OF ACTION

ACTS ON DARK SPOT PREVENTION AND CORRECTION



Gingerol and Tyrosinase Inhibition

Inhibitory effect of [6]-gingerol on melanogenesis in B16F10 melanoma cells and a possible mechanism of action

Huey-Chun Huang¹, Shao-Hua Chiu, Tsong-Min Chang

Affiliations + expand

PMID: 21670536 DOI: [10.1271/bbb.100851](https://doi.org/10.1271/bbb.100851)

Free article

The results revealed that [6]-gingerol effectively suppresses tyrosinase activity and decreases the amount of melanin in a dose-dependent manner.

Abstract

[6]-Gingerol is an active component of ginger that shows antipyretic and anti-inflammation activities. To find a novel skin-whitening agent, the melanogenesis inhibitory effects and action mechanisms of [6]-gingerol were investigated. In the present study, the effects of [6]-gingerol on mushroom tyrosinase, tyrosinase activity, and melanin content were determined spectrophotometrically, and the expression of tyrosinase and related proteins in B16F10 murine melanoma cells was evaluated by Western blotting. Furthermore, a possible signaling pathway involved in [6]-gingerol-mediated depigmentation was investigated by means of specific inhibitors. The results revealed that [6]-gingerol (25–100 μ M) effectively suppresses murine tyrosinase activity and decreases the amount of melanin in a dose-dependent manner. Additionally, it also effectively decreased the intracellular reactive oxygen species (ROS) level in a dose-dependent pattern in the same dose range. Our results indicate that [6]-gingerol inhibits melanogenesis of B16F10 melanoma and can function as a good skinwhitening agent.

<https://pubmed.ncbi.nlm.nih.gov/21670536/>



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[8]-Gingerol inhibits melanogenesis in murine melanoma cells through down-regulation of the MAPK and PKA signal pathways



Huey-Chun Huang^a, Yin-Chun Chou^b, Chia-Yin Wu^b, Tsong-Min Chang^{b,*}

^a Department of Medical Laboratory Science and Biotechnology, China Medical University, No. 91 Hsueh-Shih Road, Taichung 40402, Taiwan, ROC

^b Department of Applied Cosmetology and Master Program of Cosmetic Science, HungKuang University, No. 1018, Sec. 6, Taiwan Boulevard, Shalu District, Taichung City 43302, Taiwan, ROC

This is the first report regarding the effect of [8]-gingerol on melanin production. In the present study, it was determined that [8]-gingerol significantly inhibits tyrosinase activity and decreases melanin synthesis.

A B S T R A C T

[8]-Gingerol is an active component of Zinger and shows several pharmacological activities, such as anti-pyretic and anti-inflammation characteristics. To identify a potential skin-whitening agent, the inhibitory effects of [8]-gingerol on melanogenesis and its mechanism of action were investigated. In the present study, the effects of [8]-gingerol on mushroom tyrosinase, tyrosinase activity and melanin content were determined spectrophotometrically; the expression of melanogenesis-related proteins in B16F10 and B16F1 melanoma cells were determined by Western blotting. Furthermore, the possible signaling pathways involved in [8]-gingerol-mediated depigmentation were also investigated using specific inhibitors. The results revealed that [8]-gingerol (5–100 μ M) effectively suppressed intracellular tyrosinase activity and decreased the amount of melanin in B16F10 and B16F1 cells. In addition, [8]-gingerol also effectively decreased intracellular reactive species (RS) and reactive oxygen species (ROS) levels at the same dose range. Our results indicated that [8]-gingerol inhibited melanogenesis in B16F10 and B16F1 cells by down-regulation of both mitogen-activated protein kinases (MAPK) and protein kinase A (PKA) signaling pathways or through its antioxidant properties. Hence, [8]-gingerol could be used as an effective skin-whitening agent.

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<https://www.sciencedirect.com/science/article/pii/S0006291X13012412>

ILUMYS®

INTENSIVE ANTI-DARK SPOT

GREEN
TECH

— IN TUBO
EFFICACY

IN TUBO

DPPH TESTING

DPPH assay: Radical scavenging activity determined
by detection of DPPH radicals at 517nm

EC50 - Half Maximum Effective Concentration
It is the concentration level at which the substance is 50% effective.
The weaker the EC50, the more efficient the antioxidant

Determination of antioxidant activity

	ILUMYS®	6-gingerol	8-gingerol	10-gingerol
EC50	0.25 %	0.16 %	1.18 %	0.83 %

At 0.25%, ILUMYS® has an antioxidant activity
(equal to dry matter 0.008% or 80 ppm = good antioxidant)

**ILUMYS® has antioxidant activity:
it reduces oxidative stress involved in age spot formation**

ILUMYS®

INTENSIVE ANTI-DARK SPOT

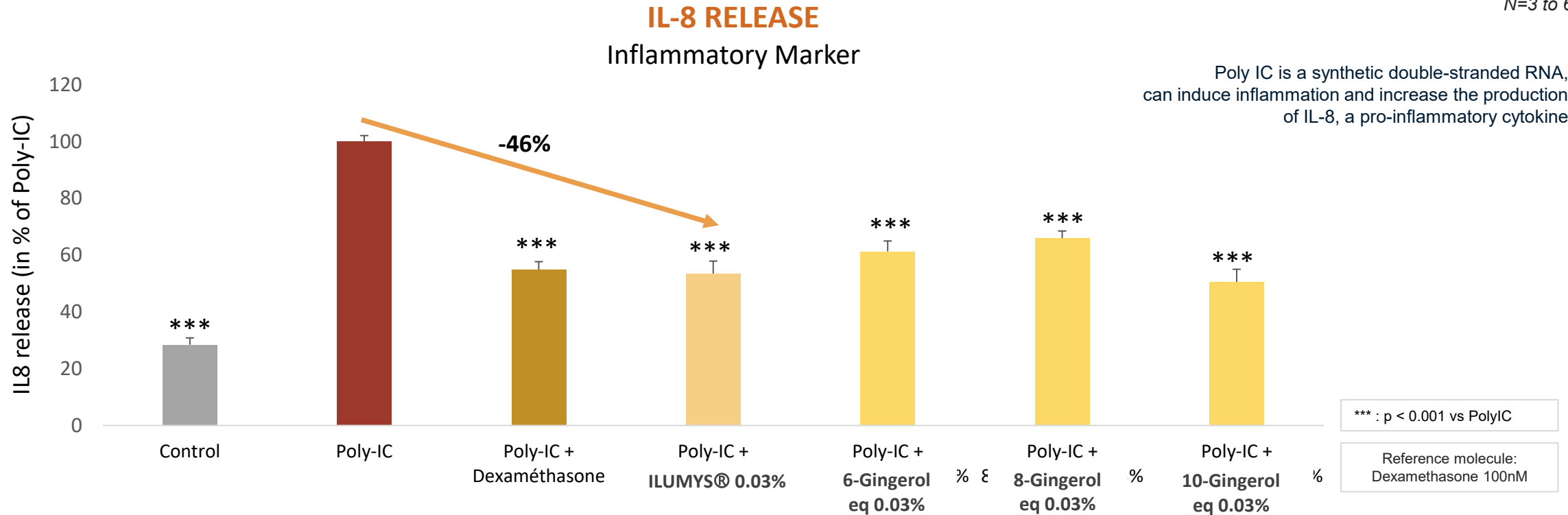
GREEN
TECH

— IN VITRO
EFFICACY

IN VITRO

ANTI-INFLAMMATION

IL-8 dosage, normalized to total protein content (BCA) on keratinocytes treated for 48 h with active ingredients, including 24 h treatment with Poly-IC stress.
N=3 to 6

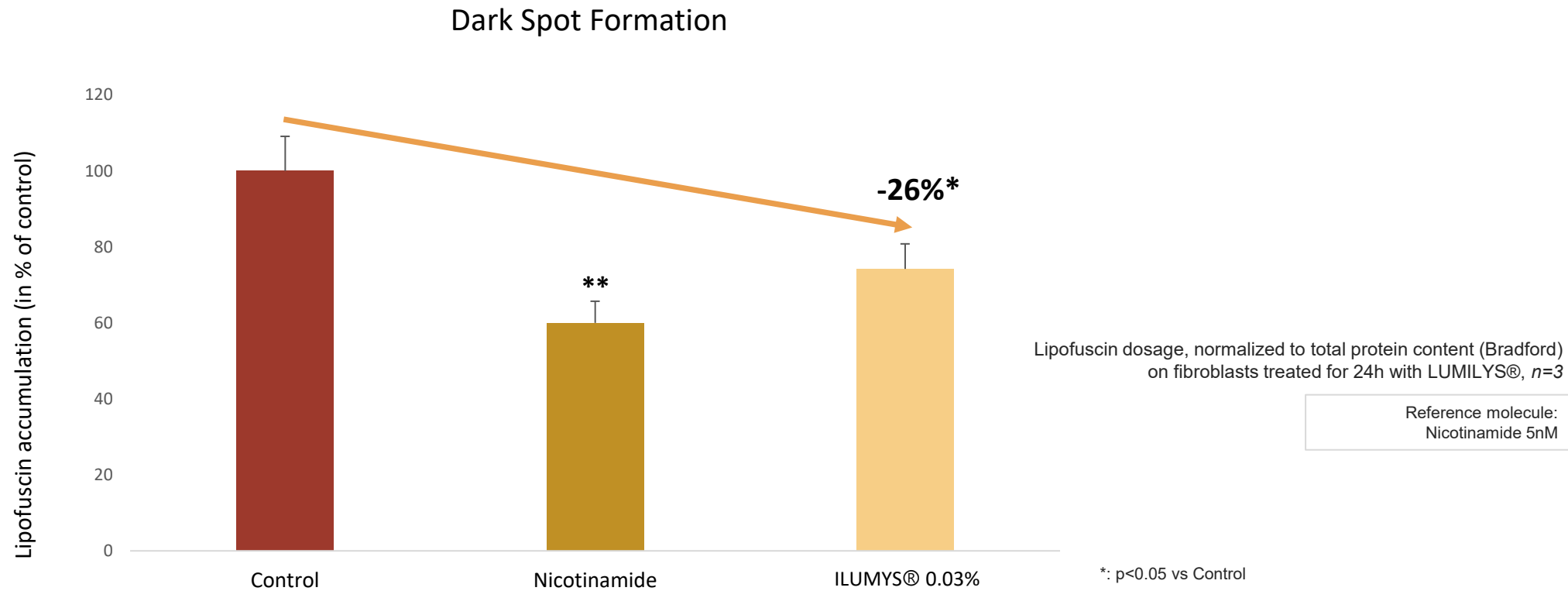


IL-8 cytokine release by keratinocytes stressed with Poly-IC

ILUMYS® reduces inflammation that participates in age spot formation

IN VITRO

LIPOFUSCIN ACCUMULATION

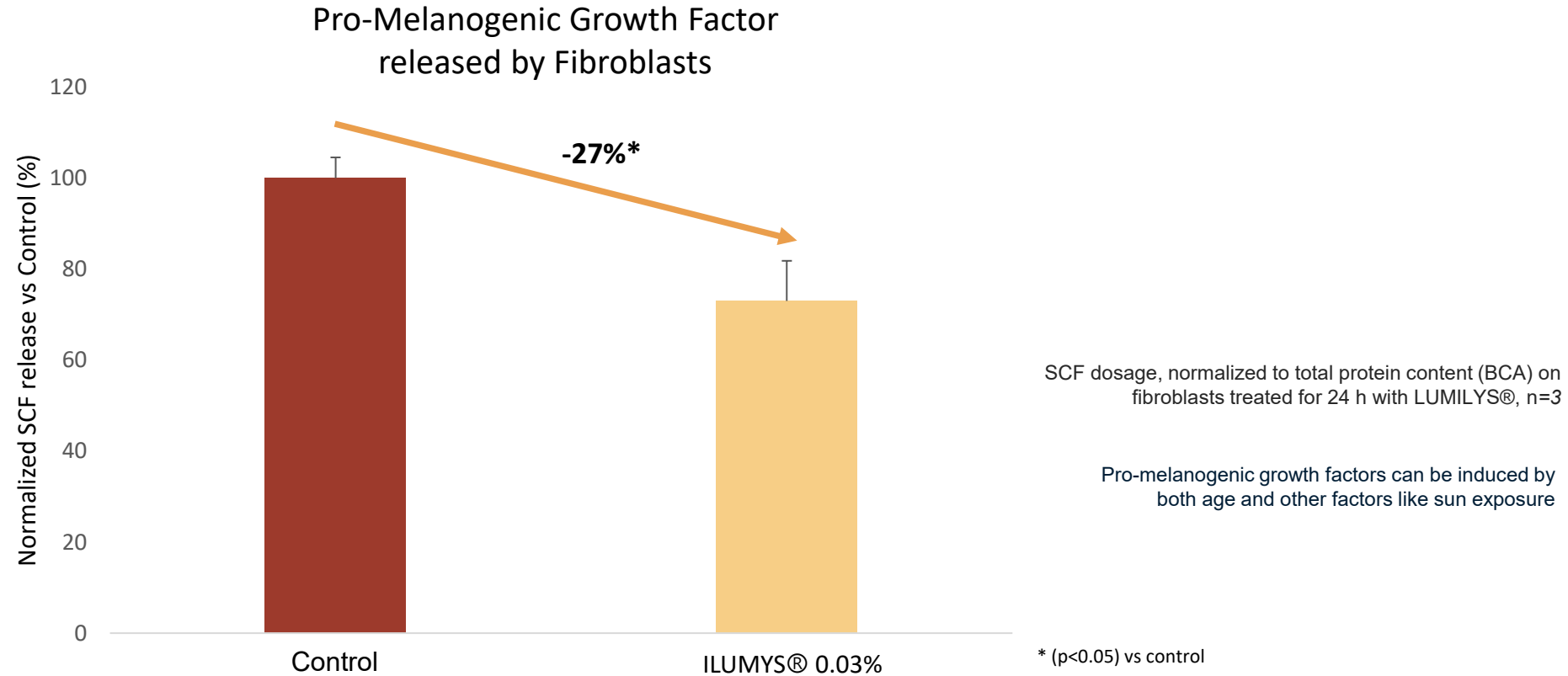


Evaluation of the effect of ILUMYS® on lipofuscin accumulation in human fibroblasts

ILUMILYS® decreases Lipofuscin accumulation

IN VITRO

STEM CELL FACTOR (SCF)



**ILUMYS® decreases the release of SCF,
a Growth Factor involved in melanogenesis upregulation.**

IN VITRO

RGMB (REPULSIVE GUIDANCE MOLECULE B)

A recent discovery:

Sensory neurons release RGMB, which acts as a melanocyte-stimulating factor and influences melanocyte survival, morphogenesis, and melanin production

Cell Reports

Report

Human sensory neurons modulate melanocytes through secretion of RGMB

Siu Yu A. Chow,^{1,2} Kazuki Nakayama,³ Tatsuya Osaki,^{1,2} Maki Sugiyama,³ Maiko Yamada,³ Hirotaka Takeuchi,³ and Yoshiho Ikeuchi^{1,2,4,5,*}

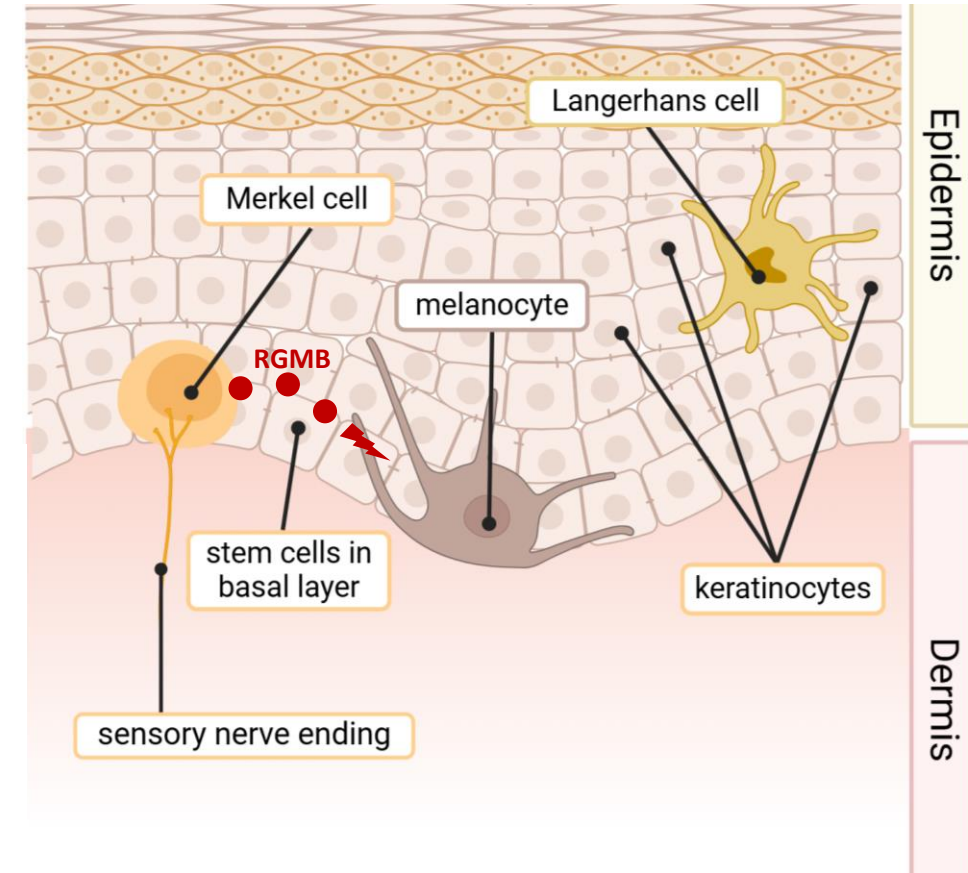
¹Institute of Industrial Science, The University of Tokyo, Tokyo, Japan

²Department of Chemistry and Biotechnology, School of Engineering, The University of Tokyo, Tokyo, Japan

³Frontier Research Center, POLA Chemical Industries, Inc., Kanagawa, Japan

⁴Institute for AI and Beyond, The University of Tokyo, Tokyo, Japan

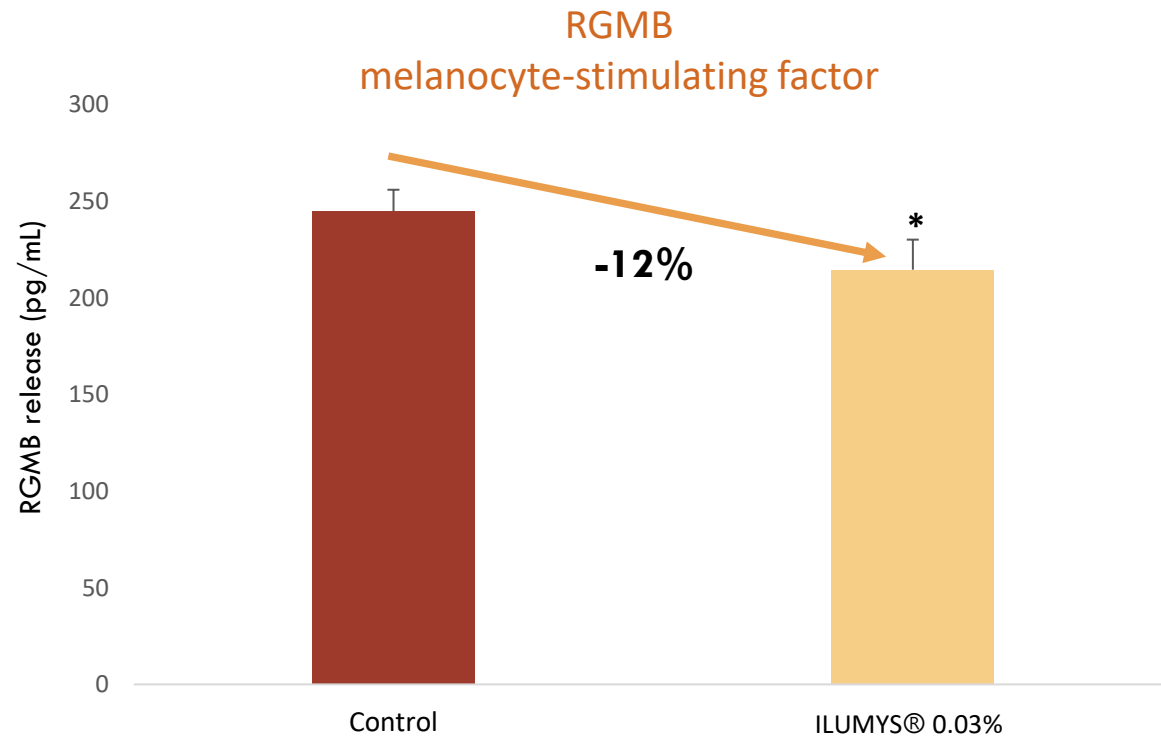
⁵Lead contact



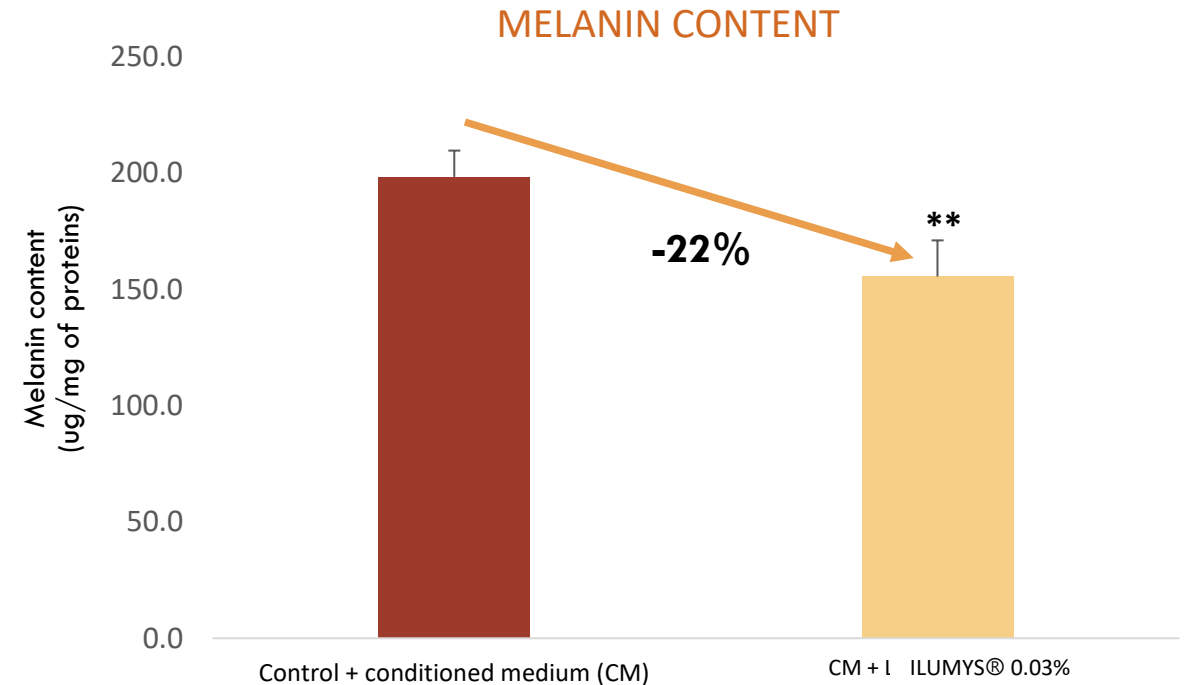
IN VITRO

RGMB INHIBITION & MELANIN SYNTHESIS

RGMB dosage in monocytes treated for 24h with ILUMYS®, n=3
Melanin dosage in melanocytes treated for 72h with conditioned media from monocytes treated with ILUMYS®, normalized to total protein content (Bradford) n=3



RGMB release by monocytes

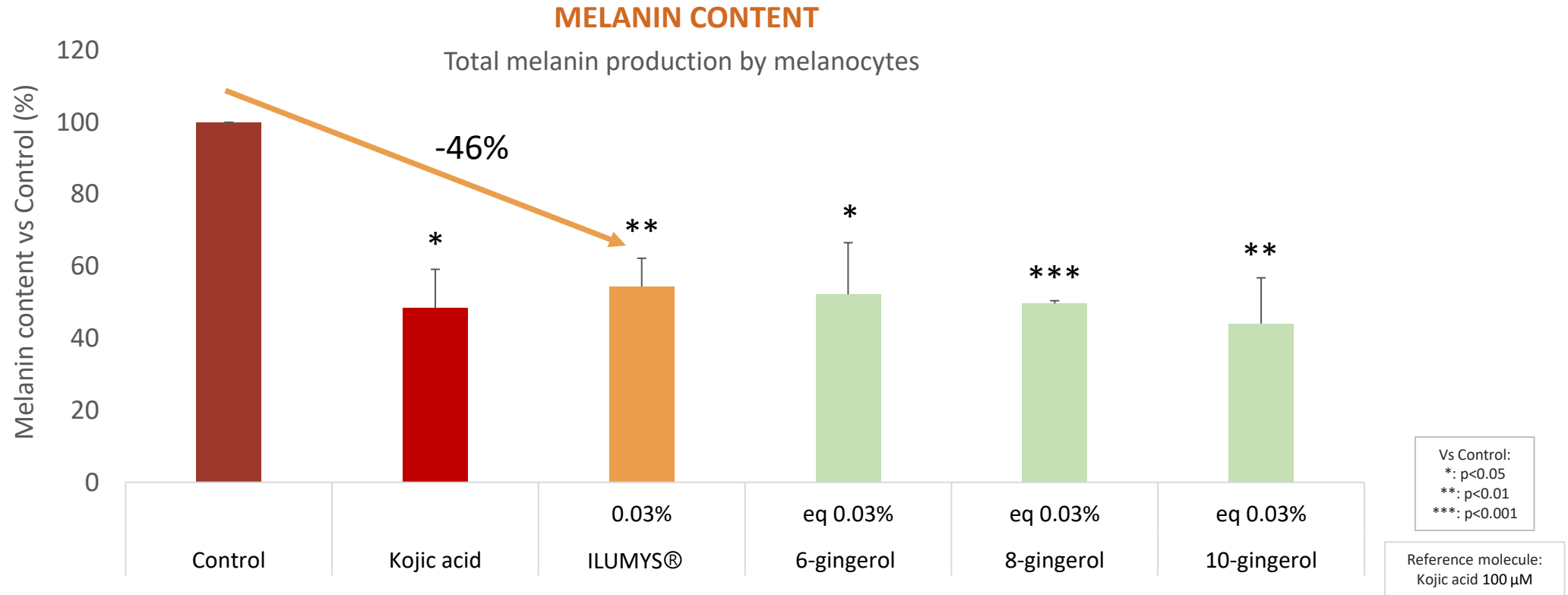


Melanin content in melanocytes treated with conditioned media from cells releasing RGMB

IN VITRO

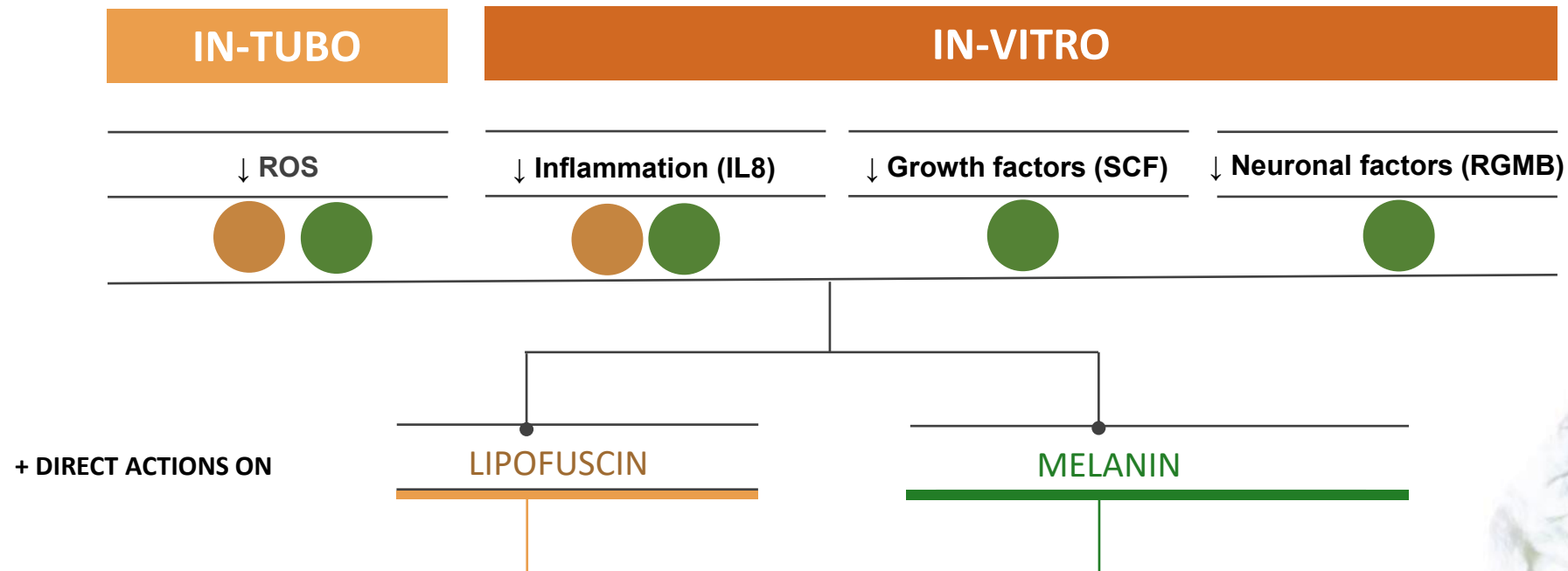
INHIBIT PIGMENTATION

Melanin content, normalized to total protein content (Lowry) in melanocytes treated for 5 days with active ingredients, n=3



ILUMYS® directly decreases melanin content in melanocytes.

SUMMARY



**MANAGE DARK SPOTS
FROM THE ROOT**

A woman's face is shown in profile, looking upwards with her eyes closed. Her skin is fair and smooth. Around her head and shoulders are vibrant, swirling paint splashes in shades of orange, yellow, pink, and blue, creating a dynamic and artistic effect. The background is a soft, light blue gradient.

ILUMYS®

INTENSIVE ANTI-DARK SPOT

GREEN TECH

— IN VIVO EFFICACY

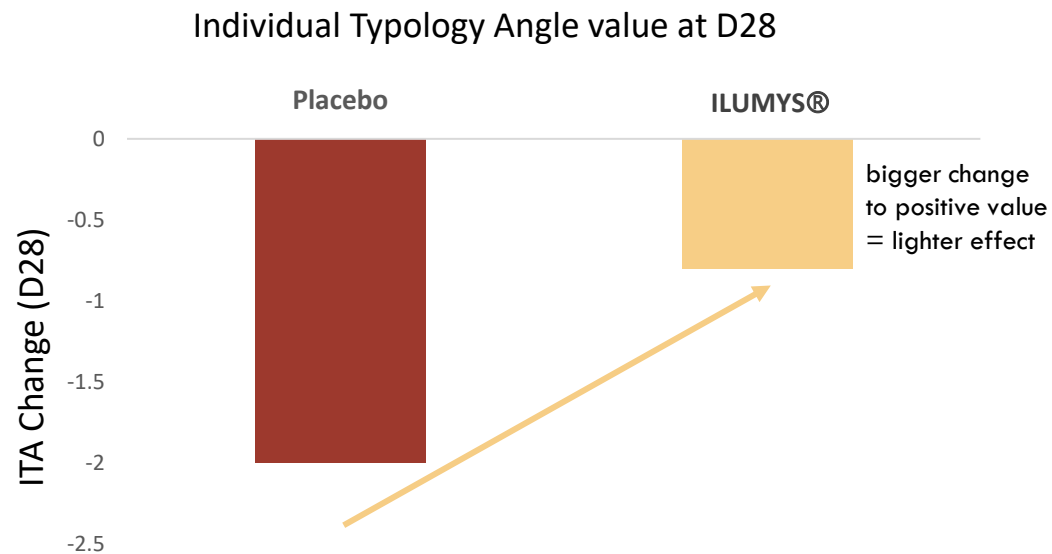
Ongoing CLINICAL STUDY:
30 Chinese volunteers
Split-face application

IN VIVO

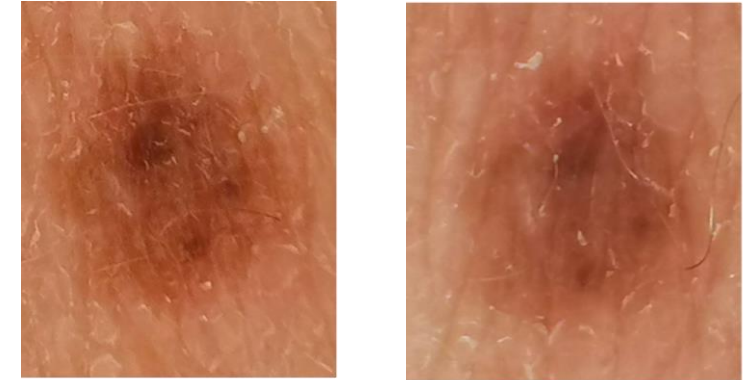
LIGHTENING EFFECT ON DARK SPOT

Clinical study under dermatological supervision:

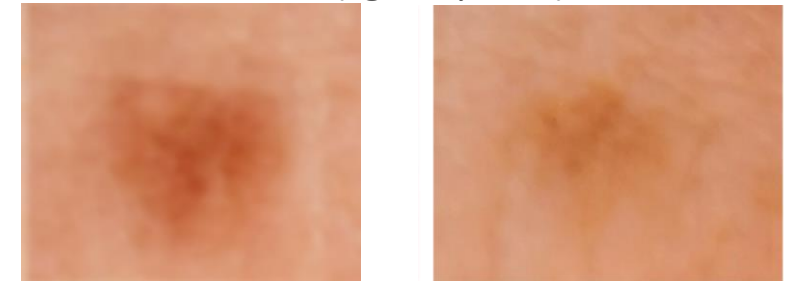
- 9 volunteers with age/dark spots
- 30-60 years old
- ILUMYS® 0.5% vs placebo
- Application: 2x/days – 28 days
- Colorimeter measurement
- ITA: individual typology angle



Volunteer 4 (age 50 yrs old)



Volunteer 8 (age 60 yrs old)



D0

D28

ILUMYS® 0.5%

ILUMYS® gives lighter effect on dark spots

PRODUCT SUMMARY

INCI name

Zingiber officinale (ginger) root extract, Triheptanoin

Preservative: None

Natural origin content (ISO 16128): 100%



COSMOS
APPROVED



IECIC
Listed

Formulation*

Oil-soluble - Recommended use: 0.5%

Cosmetic uses – Consumer targets

Radiance booster & skin lightening products

Intensive anti-dark spot / melasma

Sun care : anti-hyperpigmentation

Toxicological data

Cutaneous irritation (Patch test: at 2.5%): Non-irritating

Sensitization (HRIPT; at 2.5%): Pending

Eye irritation (HETCAM-CFIO; at 2.5%): Practically non-irritant

Mutagenicity test (Ames test - OECD 471; up to 5%): Not mutagenic

Phototoxicity (In vitro test - OECD 432): Not phototoxic

*Formulation guidelines & stability studies available on demand